

Linear algebra

Permutations: each of possible ways in which a set can be ordered.

e.g.

123, 132, 213, 231, 312, 321

recursive algorithm

- pick each element successively to the first position then place all the permutation of the rest after it.

Levi - Civita symbol:

$$\epsilon_{a_1, a_2, \dots, a_n} = \begin{cases} +1 & \text{if } (a_1, a_2, \dots, a_n) \text{ is an even perm. of } \{1, 2, \dots, n\} \\ -1 & \text{if } (a_1, a_2, \dots, a_n) \text{ is an odd perm. of } \{1, 2, \dots, n\} \\ 0 & \text{otherwise (there are repeating elements)} \end{cases}$$

Count swaps

Matrix multiplication

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & & & a_{mn} \end{pmatrix}$$

$$(A \cdot v)_i = \sum_{j=1}^n A_{ij} v_j$$

$$(AB)_{ij} = C_{ij} = \sum_{k=1}^n a_{ik} b_{kj} \quad \left. \begin{array}{l} A: n \times m \\ B: p \times n \end{array} \right\} C: p \times m$$

inner loops

Int introduction to numerical
methods - 4 -

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Linear and affine operations

Linear operation: $\underline{A}: U \rightarrow V$

$$\underline{A}(\alpha \underline{u}) = \alpha \cdot \underline{A}\underline{u}$$

$$\underline{A}(\underline{u} + \underline{v}) = \underline{A}\underline{u} + \underline{A}\underline{v}$$

If U and V are vector spaces then $\underline{A}$ is a
matrix, here we will use $U = V = \mathbb{R}^n$

Examples 2D

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{pmatrix}$$

- Scaling: $\underline{A} = \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix}$

$$\underline{A}\underline{x} = \begin{pmatrix} \alpha x_1 \\ \beta x_2 \end{pmatrix} \quad |\underline{A}| = \text{change in area of obj.}$$

- Projection: $\underline{t} = \begin{pmatrix} t_1 \\ t_2 \end{pmatrix}, |\underline{t}| = 1, \underline{P} = \underline{t} \circ \underline{t} = \begin{pmatrix} t_1 t_1 & t_1 t_2 \\ t_1 t_2 & t_2 t_2 \end{pmatrix}$

e.g. $\underline{t} = \underline{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow \underline{P} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \underline{P} \cdot \underline{x} = \begin{pmatrix} x_1 \\ 0 \end{pmatrix}$

$$\underline{t} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \underline{P} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\underline{P} \cdot \underline{x} = \frac{1}{2} \begin{pmatrix} x_1 - x_2 \\ x_2 - x_1 \end{pmatrix}$$

$$\det(\underline{P}) = |\underline{P}| = 0$$

Rotation

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad |R| = 1$$

Combination: multiplication (order matters)

e.g. $R(\frac{\pi}{2})$, $T = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$$

Affine transformation:

- translation: $\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x + t_1 \\ y + t_2 \end{pmatrix}$

add a new row:

$$\begin{pmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ d \end{pmatrix} = \begin{pmatrix} x + t_1 d \\ y + t_2 d \\ d \end{pmatrix}$$

$$d=1$$

this can also be a linear operator

often: $(A|t) \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_{11}x + a_{12}y + t_1 \\ a_{21}x + a_{22}y + t_2 \end{pmatrix}$

Now one can set the matrix by giving 3 point pairs

3.2 equations for
3.2 unknowns

